Energy efficiency

It is possible to look at how well an energy system is operating by calculating its efficiency. The efficiency of an energy transformation is a measure of how much the input energy appears as useful as the output energy.

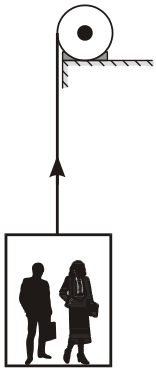
This can be calculated using the following equation:

$$\eta = \frac{E_{out}}{E_{in}}$$

$$\textbf{Efficiency} = \frac{\textbf{Useful energy output}}{\textbf{Total energy input}}$$

Note – **η** is the ratio of output to input energy. *This can never be greater than 1.* In order to convert efficiency (**η**) to a percentage, it has to be multiplied by 100.

Example



An electric lift rated at 110V, 30A raises a 700kg load to a height of 20m in 2 minutes. By considering the electrical energy input and the potential energy gained by the mass, determine the percentage efficiency of this energy transformation.

Remember – translate the time into seconds! (2 mins x 60s = 120s)

The energy going into the system is electrical:

$E_e = IVt$

$= 30 \times 110 \times 120$

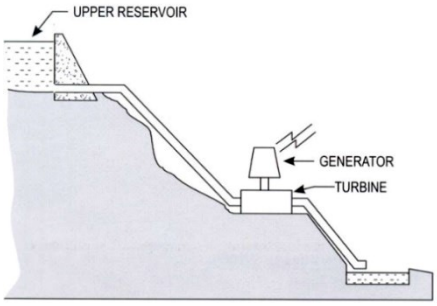
The output Energy is potential so:

$E_P = mgh$

$= 700 \times 9.8 \times 20$

$$\eta = \frac{E\ out}{E\ in} = \frac{\text{potential energy}}{\text{electrical energy}} = \frac{137.2}{396} = 0.346$$

Percentage efficiency = 0.346 × 100 = 34.6%



Q1. In a hydroelectric energy generating station, water is allowed to flow downhill through a turbine, which is connected to a generator. The water falls through a vertical height of 500m at a rate of 5000kg/s. If the energy transfer is 65% efficient, determine the amount of potential energy produced per second.

h = 500m Efficiency, η = 65% = 0.65 g = 9.8 N/kg or 9.8 ms^{-2}

= 24,500,000J = 24.5 MJ

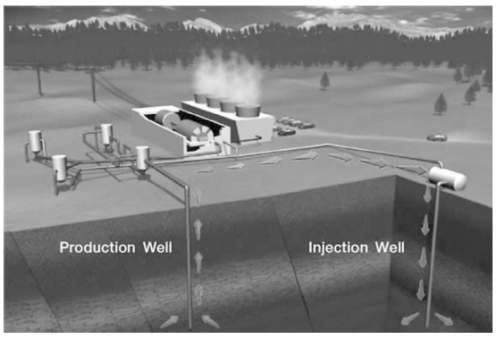
$\eta = E_{out} / E_{in}$

$E_{out} = E_{in} \times \eta$

E_{out} = 24,500,000 × 0.65

E_{out} = 15,925,000 J = **16MJ** to 2 significant figures

Q2. A Geothermal power plant uses the heat in the earth to help produce electricity.



(a) Calculate the heat energy absorbed by 100 litres of water which is pumped into the earth at 10 °C and comes out as steam at 240 °C. (1 litre of water has a mass of 1 kg.)

$\Delta T = T_2 - T_1 = 240 - 10 = 230°C$

(b) For every 15 MJ of heat energy that comes from the ground, the power plant produces 5.34 MJ of electricity. Calculate the efficiency of the power plant.

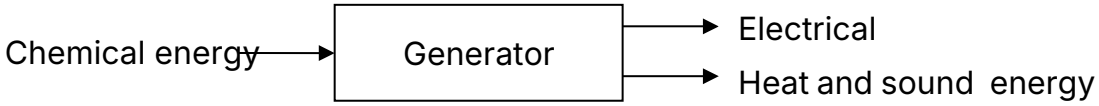
$\eta = E_{out} / E_{in}$

η = 5.34 / 15

η = 0.356 = 35.6% = 36% to 2 significant figures (remember to get a %, multiply by 100)

Energy audits

During energy transformations all the energy that goes *in* the system must come *out*. It is not possible to 'destroy' energy – it must go somewhere. Not all energy going into a system comes out as useful energy – this is called **waste energy**, or ***lost energy***. An example of this is that a light bulb turns electric energy into light energy, but it also creates heat – heat is the lost energy. Since we know that the total energy in any closed system must be constant, we can carry out meaningful calculations if we remember to take all types of input and output energy into account.



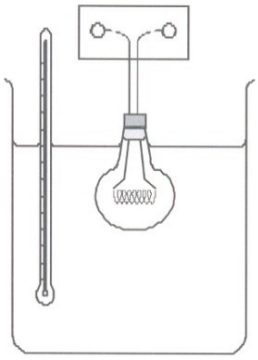
In the generator example above:

- the input energy is in the form of kinetic energy.
- the total output energy will be electrical energy, plus heat energy.

Hence, through conservation of energy $E_k = E_e + E_h$

In order to ensure that we have taken all energies into account, it is useful to carry out an **energy audit**. An energy audit is a list of all energies coming in and going out of a system. The total for the energies in ***must*** be the same as the totals for the energies out.

Worked example



In order to estimate the efficiency of an old filament electric light bulb, the bulb is immersed into a beaker of water as shown. Assuming all the heat energy is transferred to the water, it is possible to work out the efficiency of the bulb as a light producer.

Data:

Energy type going into the system is electrical (E_e) Power supply = 12 V

Current Drawn by the bulb = 5 A Volume of water in beaker = 0.5L

Initial temperature of water = 18°C Temperature of water after 10 minutes = 30°C

Time lightbulb is immersed for 10 minutes

Calculations:

t = 10 minutes x 60 seconds = 600s

Energy in (E_e) = IVt

= 5 × 12 × 600

= 36,000 J = **36 kJ**

Total Energy out is in the form of heat (E_h) and light (E_L)

$\Delta T = 30 - 18 = 12°C$

E_h out = $cm\Delta T$

= 4180 × 0.5 × 12

= 25,080 J = 25.1 kJ

The specific heat capacity (c) for water is 4180Jkg^{-1}K^{-1}

(This is also in the data booklet)

Useful energy out must equal the amount of energy in. However minus potential energy losses.

Energy in = 36 kJ Total energy out = 25.1KJ

Useful energy output = Energy out – any losses ($E_{heat\ loss}$)

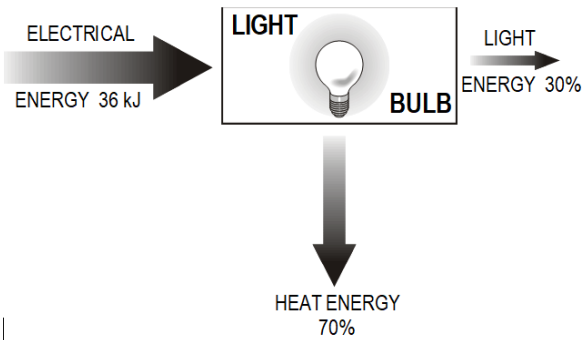
Therefore useful energy output = 36 – 25.1 = 10.9 kJ

Efficiency = Useful energy output = 10.9 x 100% = 30% (of which 30% is only converted to light)

Total energy in 36

Waste energy (heat energy) 70% of input = 25.1kJ

This can be represented by a systems loss diagram



Q3. A deep fat fryer uses 2kJ of electrical energy **per second**. It can raise the temperature of 1.5kg of cooking oil from 20°C to 80°C in 6 minutes. If the specific heat capacity of the oil is 2400J/kgK, calculate the heat energy taken by the oil every second and hence the efficiency of the fryer.

T_1 = 20 °C T_2 = 80 °C c_{oil} = 2,400J/kgK Time = 6 mins = 6 × 60 = 360s

m = 1.5kg

E_e = 2000J per second

ΔT = 80 – 20 = 60 °C

E_h = **216,000J**

$E_{e\ in} = E_e \times t = 2000 \times 360 = 720\ 000J$

$\eta = E_{out} / E_{in}$

η = 216,000 / 720,000

η = 0.3 = **30%**

Q4. When full, a container of water holds 1.5 litres of water. The initial temperature of the cold water is 20 ºC. A heater that operates from a 240V supply draws a current of 8A and has an efficiency of 75%. Determine how long it will take to heat the container of water to 90 ºC.

$C = 4180 Jkg^{-1}k^{-1}$ $\Delta T = 90 - 20 = 70$ ºC $\eta = 75\% = 0.75$

E_h = **438,900J** (Electrical energy out)

Original equation of Efficiency, $\eta = E_{out} / E_{in}$

Therefore, Electrical energy (E_e) = E_h / η = 438,900 / 0.75 = 585,200J (original energy in)

What other knowns do I have in the question?

I = 8A and V= 240 V

P = IV

P = 8 × 240 = 1,920W

Original equation E_e= IVt

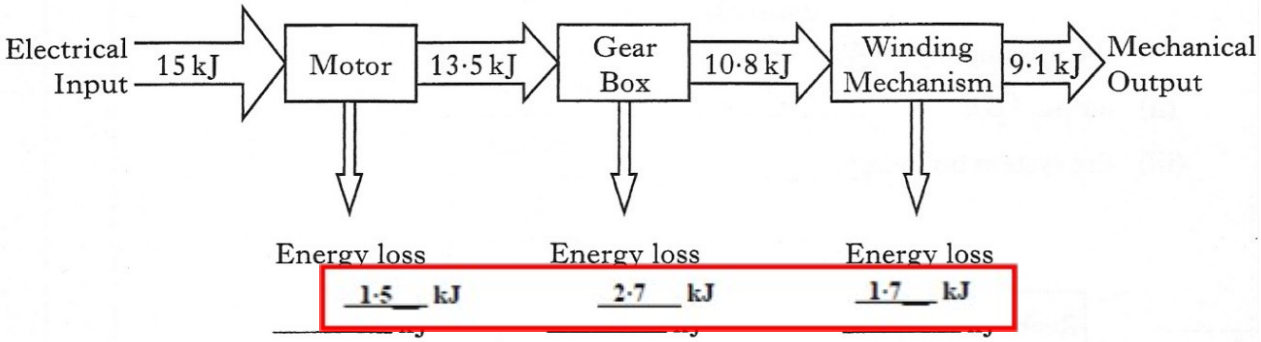
585 200 = 1920 x t

Therefore, equation re-arranged: $t = E_e / IV$

t = 585 200 / 1920 = **304.8 s** but convert to minutes = 304.8 / 60 = **5 minutes and 5 seconds**

Q5. An electric motor is used to drive a ski tow.

(*a*) Complete the energy audit below.



(*b*) Calculate the **total** energy **loss** in the ski tow.

Total energy loss = 1.5 + 2.7 + 1.7 = **5.9kJ** kJ

(*c*) Complete the following sentence using the list of phrases given.

Input energy *Lost energy* *Output energy*

"Because the input energy is always greater than the output

the efficiency will always be less than 100%."

d) How efficient is the ski tow?

$\eta = E_{out} / E_{in}$

$\eta = 9.1 / 15$

$\eta = 0.606666$ = **61%** to 2 sig. fig.

Q6. A coffee machine heats one cup of water (0.14kg) from 18 °C to 89 °C in 16 seconds. The heating element operates from a 230V, 22A supply.

a) Calculate the electrical energy going into the coffee machine.

$E_e = IVt$

$E_e = 22 \times 230 \times 16$

$E_e = 80{,}960J$ = **81kJ** to 2 sig fig.

b) Calculate the heat energy transferred to the water.

$E_h = cm\Delta T$

$E_h = 4{,}180 \times 0.14 \times 71$

$E_h = 41{,}549.2J =$ **42kJ** to 2 sig. fig.

c) Calculate the efficiency of the machine.

$\eta = E_{out} / E_{in}$

$\eta = 41{,}549.2 / 80{,}960$

$\eta = 0.5132065 =$ **51%** to 2 sig. fig.